Physical and Chemical Restraints (an Update)

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KEYWORDS

• Physical restraint • Chemical restraint • Workplace violence • Safety • Sedation

KEY POINTS

- Safety of patients and staff is a priority for any emergency department.
- The use of physical or chemical restraints may need to be used when challenged by a combative, agitated, or cognitively impaired patient who poses a threat to self or others.
- Physical or chemical restraints should be used in the most safe and ethical manner when indicated.

INTRODUCTION

Emergency department (ED) staff have the challenge of triaging, evaluating, diagnosing, treating and providing safe, ethical, and competent care to all who present to the department. Patients with altered mental status, agitation, and aggression add complexity to an already stressful and chaotic environment that is open 24 hours each day. The etiology of altered mental status may be multifactorial and include but not limited to medical illness, trauma, psychiatric disorders, cognitive dysfunction and drug use. These patients may elicit behavior that put them at risk of harming themselves, other patients, staff, or causing significant damage to the surrounding environment.[1–4] Abnormal behavior can be an obstacle to providing the medical care needed, and when de-escalation strategies fail, then restraint modalities need to be considered.[5] Hospital and departmental protocols should be written and updated as needed based on standards of care as well as the policies and regulations from the Centers for Medicare and Medicaid Services (CMS) and The Joint Commission (TJC).[6,7] The American College of Emergency Physicians (ACEP) also provides guidelines for the management of agitated patients and appropriate use of restraints in alignment with these regulatory agencies.[8] De-escalation techniques, physical and chemical restraints, and special circumstances presented by the pediatric and geriatric populations are discussed.

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EPIDEMIOLOGY

Violence is prevalent in our society, and health care workers are at a high risk of workplace violence. According to the Bureau of Labor Statistics in 2016, health care workers sustained 70% of the workplace nonfatal injuries in private industries, and nearly 20% required 3 to 5 days away from work and 21% required 31 or more days away from work to recover.[9] Workplace violence and injuries are often underreported, especially in the ED environment. High stress, fragmented communication, long waiting times, overcrowding, staff shortages, financial issues, and confusion can provoke violent behavior in predisposed individuals.[1,10–12] A recent study also showed most emergency medical services (EMS) providers reported experiencing assault at some point in their line of duty.[13]

PATHOPHYSIOLOGY

It is helpful for the emergency physician to be familiar with the characteristics that predispose patients to violent or aggressive behavior in the ED.[14] Positive predictors of potential violent behavior are male gender, having a prior history of violence, arriving in the custody of the police, being a victim of violence, and having a history of alcohol or drug abuse.[15,16] Psychiatric illness (especially schizophrenia, personality disorders, mania, and psychotic depression) also has been identified as a significant risk for violent behavior in the ED.[15] Agitation can have an organic cause, such as delirium, dementia, traumatic brain injury, neurologic disease, metabolic disturbance, infection, vascular or circulatory issues, renal or hepatic insufficiency, and endocrine dysfunction. Intoxication or withdrawal from alcohol or other recreational drugs can lead to out-of-control behavior. It is important to differentiate between functional and organic causes for a patient's agitated behavior and carefully rule out organic causes even in patients with a history of psychiatric illness or substance abuse problems. When ruling out an organic cause, a medical workup should be performed as clinically indicated.[17,18]

RECOGNIZING AND PREPARING FOR THE VIOLENT PATIENT

Violence is not usually the initial response of a patient in the ED, but is the result of increasing tension and frustration. In addition to the risks mentioned earlier, certain cues from the patient can predict the possibility of violent behavior. If these cues are noticed, a situation may be diffused with de-escalation techniques. The American Association for Emergency Psychiatry has promoted Project BETA (Best Practices in Evaluation and Treatment of Agitation) to help organizations understand that changes in culture, training, environment, resources, behavioral consulting teams, and other de-escalation strategies may lead to a decrease in use of restraints and seclusion.[19–21] This would lead to improvements in safety, length of stay, medical costs, patient-physician relationship, and enhance the overall patient experience.

When EMS is transporting a violent patient to a hospital, ideally a call should be made to alert ED staff to prepare and mobilize appropriate resources. Having response teams, adequate resources, and properly trained staff, as well as using a multidisciplinary simulation training will aid in the immediate proper management of these challenging patients.[8,22–24] Furthermore, CMS requires that facilities provide staff with training to deal with a combative, uncooperative patient.[6,7] These situations are stressful to staff and it is important to have debriefing sessions with the staff after the event.[19] Proper documentation from the time of patient arrival through discharge should identify team members, de-escalation strategies attempted, reason for use of

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restraints, and any adverse events.[6–8,25] TJC also requires all deaths related to use of restraints be reported within 24 hours.[26]

DE-ESCALATION

Before ordering any physical or chemical restraints for the agitated patient, de-escalation strategies should be attempted first, including offering reassurance, simplifying communication, demonstrating empathy, ignoring challenging arguments, adjusting the environment with dim lights, offering oral medication, and even offering food.[21] When trying to de-escalate the agitated patient, it is paramount to prioritize the safety of the patient and the treating staff, assist the patient with managing emotions, and avoid coercive interventions that may exacerbate agitation.[27,28] It may be necessary to have hospital security or police nearby in case de-escalation techniques are not successful and provider needs to implement physical and/or chemical restraints.

A previous review found that cultural bias, role perceptions, and attitude by staff were associated with the frequency of seclusion and restraint use.[19] As our country becomes more diverse, these are factors that need to be considered in using de-escalation strategies and then proper use of physical or chemical restraints if de-escalation fails.

PHYSICAL RESTRAINTS

When verbal and nonverbal de-escalation modifications have failed and the patient continues to be a threat to self or others, or their agitation prevents delivery of medical care needed, the use of physical restraints may need to be implemented. Risk factors associated with use of physical restraints in the ED include arrival by prehospital staff already in restraints, brought in by law enforcement, bipolar mania or psychosis, and insight impairment.[29]

In 2 recent studies, patients needing restraints during their ED visit were most often young patients under the influence of psychoactive substances who were eventually discharged, and confused older patients who were eventually hospitalized.[30,31]

The decision to order physical restraints is difficult because it can be perceived as coercion or punishment instead of a last resort to ensure safety, and complications may occur.[19,32,33]

ACEP has established the principles shown in **Box 1** regarding physical restraint use.[34]

Although restraining patients can be risky, appropriate use of restraints may be necessary rather than allowing the patient's out-of-control behavior to progress, which may lead to delayed treatment of the patient and/or injury to self, ED staff or others.

Every institution should have a detailed protocol for the use of restraints.[7] This protocol should identify the members of a restraint team and how to activate the team. All members of the restraint team should be appropriately trained and familiar with the restraining procedure. The team should have at least 5 members, including a designated leader who explains the process to the patient in clear language. Although not always possible, every attempt should be made to have a woman on the restraint team, especially if a woman patient is being restrained.[25] This show of force of a large, organized team alone can sometimes de-escalate the patient and avoid eventual use of restraints.

Before entering a patient's room and initiating the restraining procedure, all personal belongings that could potentially be used as a weapon or cause injury should be removed, including necklaces, lanyards with IDs, stethoscopes around the neck, neckties, and hanging earrings.[35]

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Box 1
ACEP's Established Principles for Physical Restraint Use

- Use only after verbal de-escalation attempted
- Maintain privacy and dignity of patients
- Restraint should be the least restrictive
- Staff should be properly trained in usage of restraints and monitoring of patients
- Protocols should be developed on safety, monitoring, and periodic evaluations of continuing need for physical restraints
- Proper documentation should include reasons for restraints, alternative methods tried, monitoring, and periodic assessment of need for restraints
- Federal standards, rules, and laws should be followed

Data from Use of Patient Restraints. ACEP; revised 2014, approved 1991. Retrieved from https://www.acep.org/patient-care/policy-statements/use-of-patient-restraints. Accessed July 2, 2019.

During the restraining procedure, the team leader maintains the safety of the patient's head while each of the other members takes control of a preassigned extremity. Each extremity should be controlled at the major joint (elbow, knee), and the restraint should be tied to the frame of the bed, not the handrails. All restraint positions carry certain risks, and careful monitoring of the restrained patient is essential. If aspiration is a risk, or the patent is pregnant, the patient should be restrained on the left lateral side, otherwise restraint in the supine position may facilitate the medical examination and be more comfortable for the patient. Sentinel event tracking by TJC indicates that restraining patients on their side reduces adverse restraint events.[7] Restraining a patient in the prone position predisposes them to suffocation and aspiration. If a patient requires restraint in the prone position for medical or significant safety issues, specific attention should be paid to the patient's airway and breathing. The airway must remain unobstructed at all times, the head be able to rotate side to side, the head of bed should be elevated if not contraindicated, and the expansion of the lungs must not be restricted by excessive pressure on the patient's back.[36] If 4-point restraints are used, the arms should be restrained so that one arm is up and one down. This position makes it difficult for the patient to generate enough force to overturn the stretcher. If additional restraint is needed, a sheet can be placed around the chest and tied to the stretcher, taking care that the sheet is not so tight that it prevents adequate chest expansion.[1] If the patient is semi-cooperative, or if going from 4-point to 2-point restraints, the contralateral arm and leg should be restrained. Soft restraints also can be used in semi-cooperative patients, but for patients who are combative and trying to escape, leather or nylon restraints are preferred. Soft restraints can tighten and cause circulatory compromise as a patient struggles, whereas leather restraints rarely compromise distal circulation but are more difficult to cut and remove in an emergency.[37] While restrained, patients should be under continuous observation and close monitoring. Change in position is important to prevent rhabdomyolysis, pressure sores, and paresthesias.

The treating physician should not be involved in the actual restraint process to preserve the doctor-patient relationship.[4] However, a member of the medical team should be present during the restraining process to monitor the patient. The order for implementation of restraints when a threat of imminent harm to self or staff is evident can

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be initiated by a physician or other licensed independent practitioner, or as a verbal order by a nurse; however, the physician is required to conduct a face-to-face evaluation of the patient within an hour.[26] Orders must be rewritten, for they are time limited as follows: 4 hours for adults, 2 hours for children and adolescents ages 9 to 17, and 1 hour for patients younger than 9 years. The restraint procedure requires careful documentation by the physician and the nurse. Physician documentation should cite that verbal techniques failed to calm the patient, the specific indication for the restraining procedure, the time the restraints were applied, the time-limited duration of the restraining procedure, the planned medical workup or treatment, and the patient's decision-making capacity.[33] Nursing documentation should include frequent assessment of the patient's vital signs, nutrition and hydration status, circulation, range-of-motion of extremities, hygiene and elimination, physical and psychological status, comfort, and readiness for discontinuation of restraints. The patient must also be reassessed by a physician at the end of predefined time limits. When clinically appropriate, restraints should be removed 1 at a time in 5-minute intervals until 2 are left. If the patient remains cooperative, the last 2 can be removed (**Box 2**).

TJC lists patient injury or death under restraint as 1 of the top 10 sentinel events in the ED.[7] There are many issues that can lead to patient injury or death from use of restraints, including miscommunication among staff or between staff and patient, procedural noncompliance, inadequate patient assessment, and restraint of a patient in a room not under continuous observation. During TJC inspections, failure to document and practice safe use of restraints is one of the leading citations for hospitals.[7] Any complication from the use of physical restraints should be documented, with abrasions and bruising being the most common complications.[38] Positional asphyxia has been reported to occur if the patient is restrained in the prone position.[39] Death can occur if the restraints are not applied properly or the patient is not carefully

Box 2
The Joint Commission Standards on Restraint and Seclusion

1. The [organization] uses restraint or seclusion only when it can be clinically justified or when warranted by patient behavior that threatens the physical safety of the patient, staff, or others.
2. The [organization] uses restraint or seclusion safely.
3. The [organization] initiates restraint or seclusion based on an individual order.
4. The [organization] monitors patients who are restrained or secluded.
5. The [organization] has written policies and procedures that guide the use of restraint or seclusion.
6. The [organization] evaluates and reevaluates the patient who is restrained or secluded.
7. The [organization] continually monitors patients who are simultaneously restrained and secluded.
8. The [organization] documents the use of restraint or seclusion.
9. The [organization] trains staff to safely implement the use of restraint or seclusion.
10. The [organization] reports deaths associated with the use of restraint and seclusion.

Data from Crisis Prevention Institute. Joint Commission Standards on Restraint and Seclusion/Nonviolent Crisis Intervention Training Program. Available at: https://www.crisisprevention.com/CPI/media/Media/Resources/alignments/Joint-Commission-Restraint-Seclusion-Alignment-2011.pdf Accessed July 13, 2019; with permission.

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monitored.[36] Other known complications of restraints are deep vein thrombosis, rhabdomyolysis, hyperthermia, increased sympathetic tone with vasoconstriction, metabolic acidosis due to lactic acid release from prolonged isotonic muscle contractions, and negative psychological impact. Cardiovascular collapse from metabolic acidosis has been found in many restraint-associated deaths, especially in those with acute agitation related to sympathomimetic drug use such as cocaine and methamphetamine.[36,38–42]

Physical restraints should be used for the least amount of time needed, and reevaluation for readiness of discontinuing physical restraints should be done following predefined institutional policies. Often, these may be removed once the use of chemical restraints have been successful at calming the patient. Some experts suggest that physicians are more likely to use more pharmacologic restraints or other informal restraint methods in place of physical restraints in settings with computerized forcing functions that limit their choices or require additional documentation steps to meet an institutional regulatory requirement.[43] Careful, consistent, thorough, and standardized documentation minimizes the risk of miscommunication, adverse events, and other complications from physical restraints.[33,44]

CHEMICAL RESTRAINTS

Chemical or pharmaceutical restraints, called ''rapid tranquilization'' in older literature, should also be considered in conjunction with or in place of physical restraints. Physical restraints can be counterproductive because struggling against restraints may prevent obtaining a history or completing a thorough physical examination.[45] Chemical restraints can help gain better control of the agitated patient, allow evaluation and treatment, and even hasten removal of physical restraints.[46] Complications associated with struggling against physical restraints, such as hyperthermia, dehydration, rhabdomyolysis, or lactic acidosis, can all be minimized with the early use of chemical sedation.[45]

Antipsychotics/Neuroleptics

Because so many patients with violent behavior in the ED have an underlying psychiatric illness, antipsychotic medications, either alone or combined with benzodiazepines, are used commonly in the ED setting.[47,48] Antipsychotics have a high therapeutic index and a lack of addictive potential. This class of medication has a high affinity for the dopamine-2 receptor. Contraindications to these medications include an allergy to the class, Parkinson disease, and anticholinergic drug intoxication. Relative contraindications include pregnancy, lactation, and hypovolemia.

First-Generation Antipsychotics

Haloperidol, a butyrophenone, is classified as a first-generation antipsychotic.[49] It is easily given intramuscularly (IM) at doses of 2.5 to 10 mg. The time of onset is 2 to 6 hours for oral and 30 to 60 minutes for IM/intravenous (IV). Repeat doses can be given at 30-minute to 60-minute intervals, and the desired effect is usually obtained within 3 doses. The half-life of haloperidol is 10 to 19 hours. IV administration has not been approved by the US Food and Drug Administration (FDA), but it is frequently given via this route. Extrapyramidal syndrome (EPS) is a potential side effect of haloperidol and, in rare cases, has been reported to occur days after administration, even after only 1 dose. This risk may be reduced with the concomitant use of an anticholinergic agent or benzodiazepine.[14] Occurrence of EPS is easily treated with diphenhydramine or benztropine.[50] Haloperidol can be safely given in the same syringe as

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lorazepam, thus facilitating administration, hastening onset of sedation, and resulting in fewer EPS episodes than when haloperidol is given alone.[47] Combinations of haloperidol with a benzodiazepine have been suggested to be superior to either drug alone with no difference in adverse events.[51,52] Caution must be taken if using antipsychotics such as droperidol or haloperidol, as there is a risk of QT prolongation and resultant torsades de pointes.[53]

Neuroleptic malignant syndrome (NMS) is an idiosyncratic adverse effect that can occurs in 1% of patients receiving antipsychotics, and this is more common in those receiving a first-generation antipsychotic. NMS is recognized by autonomic instability, hyperthermia, altered mental status, and muscle rigidity. Creatine phospho-kinase levels may be elevated.[54] Treatment is supportive, with aggressive cooling measures and cessation of all neuroleptics. In some cases, dantrolene is indicated for treatment of extreme muscle rigidity.

Second-Generation Antipsychotics

Second-generation (previously known as "atypical") antipsychotic medications, have a similar efficacy to first-generation antipsychotics and a better side-effect profile.[14] Risperidone, olanzapine, quetiapine, and ziprasidone are the most commonly used atypical antipsychotics in the ED. Second-generation antipsychotics inhibit dopamine-2 and serotonin receptors and thus provide more tranquilization and less sedation. An important benefit of these atypical antipsychotics is the lower incidence of EPS from the serotonergic activity.[55] This generation of antipsychotic medications has also been found to be particularly useful in cases of acute agitation from methamphetamine use.

Risperidone (Risperdal) may be given IM or orally by elixir or a recently available disintegrating tablet. Studies have shown risperidone to be as effective as IM haloperidol and less sedating than in the acute treatment of psychosis.[50] Convincing an out-of-control patient to take risperidone orally can be challenging, but onset is rapid with the orally dissolving tablets, so this option is safer and potentially more convenient for nursing staff.

Olanzapine, available IM or in orally dissolving tablets, has been found to be comparable in efficacy to either lorazepam or haloperidol in many well-controlled studies assessing its use for the treatment of agitated elderly patients with Alzheimer disease, patients with bipolar disorder with mania, and schizophrenic patients in acute agitation.[56,57] Occurrence of dystonia and akathisia are less common than with haloperidol.[58,59] Mild hypotension is a common side effect and olanzapine has significant anticholinergic properties, which could exacerbate an agitated patient who overdosed on an anticholinergic agent such as diphenhydramine or jimsonweed. Quetiapine, similar to olanzapine, can have higher sedative effects due to high affinity for antihistaminic receptors, but is available only in oral tablets.

Ziprasidone has been approved by the FDA for the treatment of acute agitation in patients with schizophrenia and patients with bipolar disorder with mania. Studies have shown IM ziprasidone has a higher affinity to serotonin receptor compared with D2 receptors and to be superior to haloperidol with fewer extrapyramidal symptoms.

Inhaled loxapine has been studied for use in the ED because of its rapid onset for treatment of agitation, peaking within 2 minutes after administration, and has been approved for acute agitation associated with schizophrenia or bipolar I disorder. It is associated with less frequent occurrence of EPS, but may cause bronchospasm.[60] An additional benefit to this drug is that it lacks sedative effects, so one can still attempt verbal de-escalation and potentially avoid additional sedation or physical restraints.

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Benzodiazepines

This class of drugs are central nervous system depressants because they augment activity at gaba-aminobutyric acid (GABA) receptors. Because benzodiazepines enhance sedation and restore inhibition, they are particularly useful in cases of agitation secondary to acute sympathomimetic drug use, such as cocaine and methamphetamine. Lorazepam is one of the more frequently used benzodiazepines for the treatment of agitation in the ED setting. It is a favorable medication because of its rapid onset, lack of active metabolites, effectiveness in patients intoxicated with a sympathomimetic agent such as cocaine, and availability in oral, IM, and IV formulations. Of all the benzodiazepines, it is the one most reliably absorbed when administered IM, so it is especially useful in the agitated patient without IV access.[46–48] Time of onset is 30 to 60 minutes orally, 10 to 20 minutes IM, and 5 to 15 minutes IV. It has a half-life of 10 to 20 hours. It is also particularly useful in patients with alcohol dependence or cirrhosis because its inactivation is preserved in the setting of liver disease.[49] Most common side effects include sedation, confusion, nausea, and ataxia. Patients must be closely monitored for respiratory depression. It is a class D agent in pregnancy and should also be avoiding in lactating women. As mentioned earlier, studies by Battaglia and colleagues have shown that the sedative effect from the combination of lorazepam and haloperidol is superior to higher doses of either medication alone. In addition, occurrence of EPS is less common in those given lorazepam and haloperidol together compared with those given haloperidol alone.[47]

Midazolam is a water-soluble short-acting benzodiazepine that has become popular because it can be administered IM or IV, intranasal, rectal, and oral. It has a rapid time to onset (18 minutes) and a time to arousal of 30 to 120 minutes (average 82 minutes). When administered IV, midazolam may result in significant hypotension, whereas IM administration has little effect on the cardiopulmonary system.[61] Some studies confirm midazolam had similar efficacy to haloperidol or lorazepam but was superior to both in that time to arousal was significantly quicker for midazolam.[62–64] A shorter time to arousal is a useful quality in the ED in that it allows these patients' evaluations to be completed more rapidly, allowing for faster determination of the appropriate disposition of the patient. Midazolam has recently become a favorable benzodiazepine option for use in agitated patients by prehospital staff.[63]

Ketamine

Ketamine, previously used most commonly for procedural sedation in the ED, has recently become a first-line agent to treat agitation in the prehospital and ED setting.[65,66] It interacts with several receptors, including N-methyl-D-aspartate, nitric oxide synthase, and multiple opioid receptors. It has a quick onset of action (2–3 minutes) and a duration of 5 to 30 minutes.[14,66] When compared with haloperidol, in the prehospital setting, it is noted to cause a higher rate of complications, including the need for intubations.[67–69] Some have recommended it be avoided in schizophrenia, as it can exacerbate the symptoms.[20] Studies of ketamine for sedation of agitated patients in the ED are limited to date, but prehospital studies and early data in the ED are promising. Ketamine has been shown to have faster onset to sedation and no significant changes in re-dosing, or in adverse effects, compared with other agents.[66] Due to variation in dosing and use of additional sedative agents, the need for airway management and rates of intubation vary in different studies on ketamine.

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Special Populations

Restraints in the geriatric patients

In the past decade, we have seen an increase in the population 65 years and older and this growth is expected to continue for the next several decades. This group has grown by 30% since 2010, and in 2018, people older than 65 years comprised 16% of the US population.[70] In 2015, there were 47.8 million and it is projected that by 2060, this number will reach 98.2 million.[70] As this segment of the population grows, so will the number of elderly patients presenting to the ED with acute agitation and psychiatric disorders. The elderly also often have multiple comorbidities, use of polypharmacy, dementia, Alzheimer disease, and other cognitive impairments. As our country becomes more culturally and linguistically diverse, so will this segment of the population, and communication may add another challenge. There also will be an increase in alternative living arrangements (ie, nursing homes, assigned living), where historically restraints are used more often.[71]

The agitated, confused, suicidal, delusional, or aggressive geriatric patient poses a unique challenge to the ED physician and staff.[72] Studies looking at the use of restraints in the elderly presenting to the ED are limited.[31,72–74] Proper de-escalation techniques, including use of family to calm the agitated patient, should be attempted before any chemical or physical restraint is used. The use of chemical or physical restraints has been associated with prolonged length of stay in the ED for elderly patients who present with behavioral complaints, and this puts them more at risk for adverse events.[71,75] If chemical restraints are ordered, the physician needs to be cognizant that the elderly may metabolize medications differently because of age and other medical conditions and are often on multiple medications that may increase their risk of an adverse event when combined with sedatives or antipsychotics.[76] Elderly patients may already be on medications that cause QT prolongation or sedation, and the use of certain antipsychotics or anxiolytics may exacerbate symptoms and lead to serious complications. Documentation should include reasons for chemical sedation, use of screening agitation scales, as well as reason for agent and dosage chosen.[23] As with adults and children, little is known of the psychological adverse outcomes of restraint use in the geriatric population,[72] and providers should minimize duration of restraining whenever possible.

Restraints in the pediatric population

The pediatric patient may present agitated or violent as a result of a medical or psychiatric illness, psychological trauma, delirium, substance use, developmental/behavioral/cognitive disorders (attention-deficit/hyperactivity disorder, hyperactive, autism, impulsivity, intellectual disability), fear, pain, or from the stress of the ED environmental stimuli.[77,78] This population presents unique challenges to the ED staff, as they may be of nonverbal age, or not able to communicate their feelings of fear, anxiety, or even pain. They also may not comprehend explanations from the staff.[77] The number of mental health–related visits to the ED has increased, accounting for approximately 5.0% to 6.8% of all pediatric ED visits.[79,80] Although a large proportion of ED visits in the pediatric population are young adults (17–24 years old), the biggest increase from 2011 to 2015 was seen in the adolescent and nonwhite youth. There has been a 2.5-fold increase seen in self-injury and suicide-related visits in the adolescent age.[80]

Clinicians need to be aware of specific standards from CMS and TJC related to this group of patients, because they include different parameters from those used in adults requiring restraints.[81,82] The American Academy of Pediatrics and ACEP developed a joint policy statement based on federal regulations to provide

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guidelines for clinicians caring for pediatric mental health emergencies.[78] Restraints used are to be the least-restrictive, age-appropriate methods, and not as a punishment or staff convenience.[8,27] There is a paucity of rigorous studies evaluating risks, efficacy, adverse events, of verbal, chemical, or physical restraints. Staff training; use of a family member to calm the patient; use of child psychologist, social worker, or other specially trained staff; child-life specialist; environmental adaptations; verbal restraints; and age-appropriate de-escalation strategies have shown to be helpful in calming patients.[80,81,83,84] Restraint and seclusion can result in significant long-term psychological trauma, which has not been well studied.[77] The use of a child-life specialist, child psychiatrist, or mental health social worker have shown to aid in the management of pediatric patients with mental health presentation.[77,84]

PREVENTIVE MEASURES

Various environmental, administrative, and behavioral preventive measures can be implemented to help make the ED a safer environment.[85–87] Certain environmental designs can strategically minimize violence in hospitals. Monitoring systems, such as metal detectors and security cameras, have been found to discourage aggressive behavior in the ED and other industries.[88] Waiting rooms should be designed to accommodate and assist visitors and patients who may have a delay in service. The triage area, waiting rooms, and reception areas should have heavy furniture that cannot be lifted and used as weapons.

Uniformed security available 24 hours a day is one of the best ways to increase the safety of a department and hospital. Alarm systems and other ways of emergency signaling should be in place in all EDs. A panic button in patient rooms, bathrooms, and hallways can be helpful. A verbal alarm system to get the attention of other staff, such as a code like ''Dr Armstrong to room 8,'' is another effective way to mobilize appropriate staff without upsetting other patients.[33]

Administrative controls to help reduce patient violence include designing staffing patterns to minimize patient waiting times and to prevent staff from working alone. Restricting the movement of the public throughout the hospital or various sections of the ED by card-controlled ID access has also been recommended. Behavioral prevention strategies include continually educating and training all staff to recognize and manage violent events and assaults, resolve conflicts when dealing with patients, and maintain situational awareness. Training should prioritize a patient-centric approach, de-stigmatize illnesses, collaborate in management, and be a part of other institutional and societal activities that can improve the patient experience.[19,27,89]

SUMMARY

As EDs continue to become more crowded, safely managing violent or aggressive patients will continue to be challenging. Appropriately trained personnel in verbal de-escalation and behavioral/environmental modifications, adherence to protocol, careful attention to documentation, and good common sense will minimize complications and ensure safe use of physical or chemical restraints when indicated.

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